

General First Login Information and Security

MEDBRIDGE+ — ULTRA-SECURE NATIONAL AUTHENTICATION SYSTEM

Zero-Trust Healthcare Identity & Access Architecture

1. MEDBRIDGE+ SECURE LOGIN PORTAL

MEDBRIDGE+ NATIONAL HEALTH SYSTEM

"Protected Healthcare Infrastructure of the Republic of Rwanda"

ZERO-TRUST SECURE AUTHENTICATION

AUTHORIZED ACCESS ONLY

USER ROLE

[ ▼ Select Authorized Role ]

• Doctor

• Nurse

• Midwife

• Pharmacist

• Lab Technician

• Ambulance Team

• Hospital Director

• Ministry of Health

• RBC Staff

• RSSB Staff

• RIB Investigator

+-----+

PROFESSIONAL LICENSE / STAFF ID:

[ ]

NATIONAL ID NUMBER:

[ ]

OFFICIAL INSTITUTION EMAIL:

[ ]

WORKING HEALTH FACILITY:

[ ▼ Select Registered Facility ]

PASSWORD:

[ ]

SECURITY LEVEL:

☒ High Security Login

☐ Remember This Device (Restricted)

[  SECURE LOGIN ]

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SECURITY STATUS:

 Encrypted Session Active

SYSTEM STATUS:

☒ National Health Network Operational

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## 2. MULTI-FACTOR AUTHENTICATION (MFA)

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| MULTI-FACTOR SECURITY VERIFICATION |

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**STEP 1:**

☒ Password Verified

**STEP 2:**

☒ OTP SENT TO REGISTERED PHONE

**PHONE NUMBER:**

+250 788 \*\*\* \*\*

**ENTER OTP:**

**STEP 3:**

**BIOMETRIC AUTHENTICATION**

- ☒ Fingerprint Scan
- ☒ Face Recognition
- ☒ Iris Recognition (Optional High-Level Accounts)

[ VERIFY BIOMETRICS ]

**3. NATIONAL BIOMETRIC VERIFICATION**

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| RWANDA NATIONAL BIOMETRIC VALIDATION

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**BIOMETRIC STATUS:**

- ☒ Fingerprint Match Successful
- ☒ Face Match Successful

**NATIONAL ID VALIDATION:**

☒ VERIFIED

**PROFESSIONAL LICENSE STATUS:**

 **ACTIVE**

**SECURITY CLEARANCE:**

**LEVEL 4 — CLINICAL ACCESS**

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**4. DEVICE TRUST & REGISTRATION**

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| DEVICE TRUST MANAGEMENT |
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```

**DEVICE NAME:**

**Samsung Galaxy Tab S10**

**DEVICE ID:**

**MBX-DEV-889221**

**DEVICE STATUS:**

☒ **Trusted Hospital Device**

**LOCATION:**

**Kigali University Teaching Hospital**

**LAST SECURITY SCAN:**

**27 May 2026 — 07:12 AM**

**THREAT STATUS:**

 **No Threats Detected**

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**5. GEOLOCATION & FACILITY VALIDATION**

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| LOCATION & FACILITY SECURITY |

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CURRENT LOCATION:  
Kigali — Rwanda

GPS STATUS:  
☒ Active

AUTHORIZED FACILITY:  
CHUK — Kigali

NETWORK TYPE:  
☒ Hospital Secure Network  
☐ Public Network

ACCESS STATUS:  
☒ Authorized Access Granted

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6. AI THREAT DETECTION ENGINE

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| AI CYBER DEFENSE ENGINE |

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LIVE THREAT ANALYSIS:  
☒ Monitoring login behavior  
☒ Detecting abnormal activity  
☒ Monitoring failed login attempts  
☒ Checking impossible travel events  
☒ Monitoring insider threats

AI SECURITY STATUS:  
● SAFE SESSION

RISK SCORE:

LOW RISK — 2%

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7. ROLE-BASED ACCESS CONTROL (RBAC)

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| ROLE AUTHORIZATION ENGINE                               |
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```

USER:  
Dr. Claude Ndayambaje

ROLE:  
Ophthalmologist

- AUTHORIZED MODULES:
- ☒ Clinical Consultations
  - ☒ Eye Imaging
  - ☒ Prescriptions
  - ☒ Surgery Scheduling

- BLOCKED MODULES:
- ☐ Financial Systems
  - ☐ National Intelligence Data
  - ☐ RIB Investigation Logs
  - ☐ System Configuration
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8. SESSION ACTIVATION & SHIFT SECURITY

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| SECURE SHIFT ACTIVATION                               |
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```

SHIFT TYPE:

- ☒ Day Shift
- ☐ Night Shift

**WORK AREA:**

- ☒ Outpatient Department
- ☐ Inpatient Ward
- ☐ Emergency Unit
- ☐ Operating Theater

**SESSION DURATION:**

8 Hours Maximum

**AUTO-LOCK TIMER:**

2 Minutes Inactivity

[ START SECURE SESSION ]

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**9. HIGH-LEVEL GOVERNMENT ACCOUNT SECURITY**

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| GOVERNMENT & NATIONAL SECURITY ACCESS
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```

**APPLIES TO:**

- Ministry of Health
- RBC
- RSSB
- RIB
- National Cyber Defense

**ADDITIONAL SECURITY:**

- ☒ Hardware Security Key
- ☒ Smart Government Card
- ☒ Dual Approval Authentication
- ☒ AI Behavior Monitoring
- ☒ National VPN Required

**CLEARANCE LEVEL:**  
**LEVEL 7 — NATIONAL ACCESS**

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## **10. EMERGENCY ACCESS SYSTEM ("BREAK GLASS")**

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| 🚨 EMERGENCY ACCESS OVERRIDE |
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**EMERGENCY ACCESS REQUESTED:**  
Patient in life-threatening condition

**REASON:**  
Cardiac Arrest — Unknown Identity

**ACCESS TYPE:**  
Temporary Emergency Clinical Access

**SECURITY ACTIONS:**  
☒ Full Audit Logging Enabled  
☒ Supervisor Notification Sent  
☒ Session Recording Activated

**ACCESS DURATION:**  
15 Minutes

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## **11. ADVANCED PASSWORD POLICY**

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| PASSWORD SECURITY REQUIREMENTS |
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```



**PASSWORD RULES:**

- ☒ Minimum 14 Characters
- ☒ Uppercase Letters Required
- ☒ Lowercase Letters Required
- ☒ Numbers Required
- ☒ Symbols Required
- ☒ No Reused Passwords
- ☒ Auto Expiration Every 60 Days

**EXAMPLE:**

**M3dBridge#Rw2026!**

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**12. ACCOUNT LOCKOUT & ATTACK PREVENTION**

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| INTRUSION PREVENTION                               |
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**FAILED LOGIN LIMIT:**

**5 Attempts**

**AFTER 5 FAILURES:**

- ☒ Account Locked
- ☒ Security Team Alerted
- ☒ Device Blacklisted Temporarily
- ☒ AI Threat Investigation Started

**LOCKOUT TIME:**

**30 Minutes**

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**13. LIVE SECURITY AUDIT TRAIL**

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| REAL-TIME AUDIT LOGGING                               |
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**EVERY ACTION LOGGED:**

- ☒ Login Attempts
- ☒ Patient Records Viewed
- ☒ Prescriptions Issued
- ☒ Lab Results Modified
- ☒ Data Exports
- ☒ System Configuration Changes

**AUDIT STORAGE:**

Rwanda National Data Center

**TAMPER PROTECTION:**

- ☒ Blockchain Verification Enabled

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**14. NATIONAL CYBER DEFENSE INTEGRATION**

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| **NATIONAL CYBERSECURITY INTEGRATION**

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**CONNECTED TO:**

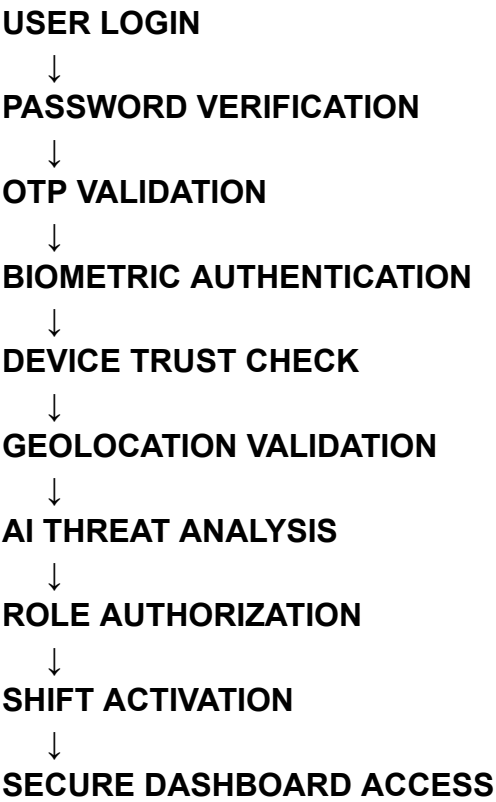
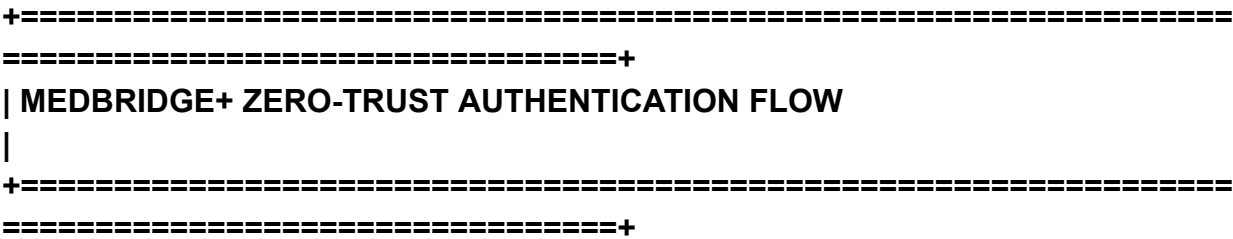
- Rwanda National Cyber Security Authority
- National Data Center
- Ministry of ICT
- Health Emergency Response Systems

**LIVE MONITORING:**

- ☒ Malware Detection
- ☒ Ransomware Protection
- ☒ DDoS Protection
- ☒ Intrusion Detection
- ☒ AI Threat Intelligence

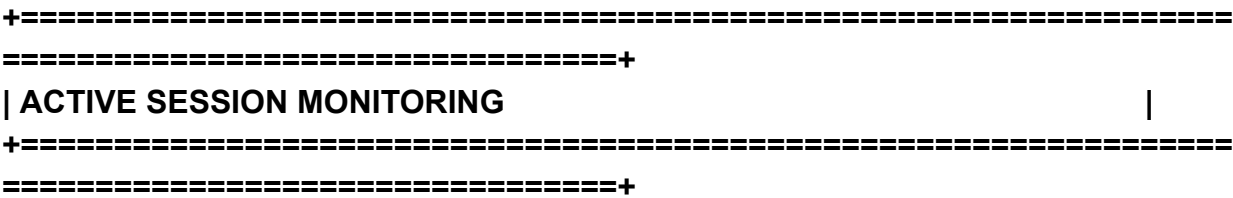
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## 15. SECURE LOGIN FLOW ARCHITECTURE



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## 16. SESSION MONITORING PANEL



**USER:**

Dr. Claude Ndayambaje

**FACILITY:**

Rwanda Military Hospital

**DEVICE:**

Hospital Tablet — Trusted

**LOGIN TIME:**

07:52 AM

**SESSION STATUS:**

 **ACTIVE**

**LAST ACTION:**

Viewed Retinal Imaging

**SECURITY SCORE:**

98 / 100

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**17. AI FRAUD & INSIDER DETECTION**

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| AI INSIDER THREAT MONITOR                                     |
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**AI MONITORS:**

- ☒ Unusual patient access
- ☒ Mass data downloads
- ☒ Unauthorized exports
- ☒ Suspicious login locations
- ☒ Privilege abuse attempts

**CURRENT STATUS:**

 **No abnormal activity detected**

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18. SECURE MOBILE ACCESS

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| MOBILE SECURITY |

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MOBILE DEVICE MANAGEMENT:

- ☒ Remote Wipe Enabled
- ☒ Device Encryption Enabled
- ☒ App Tamper Protection
- ☒ Offline Secure Cache
- ☒ Jailbreak/Root Detection

APP STATUS:

☒ Secure

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19. NATIONAL SECURITY LEVELS

| Level   | Role Type   | Security Strength  |
|---------|-------------|--------------------|
| Level 1 | Citizens    | OTP + Password     |
| Level 2 | Nurses      | MFA + Biometrics   |
| Level 3 | Doctors     | MFA + Device Trust |
| Level 4 | Specialists | AI Monitoring      |
| Level 5 | Directors   | Hardware Key       |

|         |           |                |
|---------|-----------|----------------|
| Level 6 | RBC/RSS B | Government VPN |
|---------|-----------|----------------|

|         |         |                    |
|---------|---------|--------------------|
| Level 7 | RIB/MoH | Dual Authorization |
|---------|---------|--------------------|

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## **20. FINAL SECURE AUTHENTICATION STRUCTURE**

### **AUTHENTICATION LAYERS**

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**Password Authentication**  
**OTP Verification**  
**Biometric Verification**  
**Device Trust Validation**  
**Geo-Fencing**  
**AI Threat Analysis**

### **ACCESS CONTROL**

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**Role-Based Access**  
**Department Restrictions**  
**Shift-Based Access**  
**Facility Restrictions**

### **MONITORING**

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**Live Audit Logging**  
**Blockchain Verification**  
**AI Fraud Detection**  
**National Cyber Defense**

### **EMERGENCY SECURITY**

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**Break Glass Access**  
**Supervisor Approval**  
**Temporary Access Tokens**  
**Full Session Recording**